

Awesome AI Scientist: A Survey of AI Systems for Scientific Discovery

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Awesome AI Scientist

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Abstract

This survey synthesizes 31 AI Scientist systems, products, benchmarks, and datasets into a review-style manuscript focused on taxonomy, capabilities, autonomy, evaluation, risks, and future directions.

Keywords: AI Scientist, scientific discovery agents, automated research, benchmarks, survey

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1 Introduction

The phrase *AI Scientist* increasingly refers to a family of systems that attempt to automate parts of the scientific lifecycle rather than merely answer research questions. The strongest claims come from end-to-end systems that generate ideas, write code, run experiments, analyze results, and draft papers. Adjacent systems cover narrower but still consequential slices: literature synthesis, hypothesis generation, chemistry planning, autonomous laboratories, algorithm discovery, scientific coding, reproducibility, and deep research reporting.

This survey treats the field as an ecosystem rather than a single benchmark race. The reason is practical: a system such as The AI Scientist is not comparable to Elicit, PaperBench, Coscientist, or AlphaEvolve on one scalar axis. Each artifact occupies a different position in the research stack. A useful review therefore needs a taxonomy, a timeline, an autonomy model, and an evaluation map that connects systems to the benchmarks and real-world settings used to justify them.

2 Corpus and Review Method

The review is based on 31 structured records generated from primary or official sources where possible. The corpus includes papers, repositories, product pages, benchmark pages, dataset pages, and exported PDF/Markdown artifacts. Items were included if they automate or substantially augment scientific research workflows, or if they evaluate such systems. General chatbots and unrelated foundation models were excluded unless they appeared as part of a research-agent workflow.

The resulting corpus contains: End-to-end AI Scientist systems: 3, Research workflow agents: 1, Hypothesis and research-planning systems: 1, Domain scientific-discovery agents: 6, Science agent product suites: 2, Deep-research products: 1, Literature-review products: 2, Evaluation benchmarks and datasets: 15. This distribution already reveals the maturity of the field: benchmark work is nearly as numerous as system-building work, suggesting that the community is still negotiating what counts as scientific progress by agents.

Category	Count	Interpretation
End-to-end AI Scientist systems	3	System or product layer
Research workflow agents	1	System or product layer

Category	Count	Interpretation
Hypothesis and research-planning systems	1	System or product layer
Domain scientific-discovery agents	6	System or product layer
Science agent product suites	2	System or product layer
Deep-research products	1	System or product layer
Literature-review products	2	System or product layer
Evaluation benchmarks and datasets	15	Evaluation infrastructure

3 Taxonomy of the AI Scientist Ecosystem

The ecosystem is best understood as a layered stack. At the top are end-to-end AI Scientist systems, which attempt to bind ideation, implementation, experimentation, and paper drafting into one loop. The middle layer contains domain discovery agents, such as chemistry, biology, materials, and algorithm-discovery systems. The bottom and cross-cutting layer contains research products and benchmarks: tools that either support human researchers directly or measure whether agents can perform credible research work.

3.1 End-to-end AI Scientist systems

- **DeepScientist** (2025-09-30, L4): Local-first autonomous research studio for long-horizon scientific discovery and paper-oriented workflows.
- **The AI Scientist** (2024-08-12, L4): Autonomous scientist for end-to-end machine learning paper generation.
- **The AI Scientist-v2** (2025-04-07, L4): Autonomous scientist for open-ended machine-learning discovery via agentic tree search.

3.2 Research workflow agents

- **Agent Laboratory** (2025-01-08, L3): Autonomous research assistant

3.3 Hypothesis and research-planning systems

- **AI Co-Scientist / Co-Scientist** (2025-02-19, L3): Collaborative co-scientist for scientist-in-the-loop hypothesis generation and research planning.

3.4 Domain scientific-discovery agents

- **A-Lab / GNoME Materials Discovery Lineage** (2023-11, L4): Autonomous materials discovery stack that combines large-scale learned candidate generation with robotic closed-loop synthesis and characterization.
- **AlphaEvolve** (2025-05-14, L3): A Gemini-powered evolutionary coding agent for algorithm discovery and system optimization.
- **ChemCrow** (2023-04-11, L2): A tool-augmented chemistry assistant that uses GPT-4 plus expert-designed tools to solve reasoning-intensive chemistry tasks.
- **Coscientist** (2023-12-20, L4): A GPT-4-based autonomous chemistry co-scientist for planning, executing, and optimizing laboratory experiments.
- **Robin** (2025-05-19, L3): Semi-autonomous scientific discovery system
- **SciAgents** (2024-09-09, L3): Domain-specific autonomous co-scientist for graph-grounded hypothesis generation in materials science.

3.5 Science agent product suites

- **Asta** (2025-08-26, L2): A science-focused agent ecosystem that combines research assistants, benchmark infrastructure, and developer resources.
- **FutureHouse Platform Agents** (2025-05-01, L2): Scientific agent platform

3.6 Deep-research products

- **OpenAI Deep Research** (2025-02-02, L2): An agentic deep-research product for multi-step web research, source analysis, and citation-grounded report generation.

3.7 Literature-review products

- **Consensus** (2021, L1): AI academic search and evidence-synthesis workspace for scientific research.
- **Elicit** (2021-08-31, L1): An AI research assistant and literature-review platform focused on paper search, evidence extraction, and synthesis.

This taxonomy suggests that autonomy is not a binary property. Literature-review products are often useful at L1, deep-research products and product suites sit near L2, workflow agents approach L3, and the strongest end-to-end or laboratory systems claim L4 behavior in bounded domains. No record in this corpus should be read as demonstrating unconstrained L5 scientific autonomy.

4 Historical Development

The modern ecosystem begins with tool-augmented scientific agents and autonomous laboratory work in 2023, expands into broad AI Scientist and benchmark efforts in 2024, and becomes a much

denser landscape in 2025-2026. The timeline shows a shift from individual proof-of-concept systems toward benchmark suites and productized scientific assistants.

Year	Items	Representative records
2021	2	Consensus, Elicit
2023	4	ChemCrow, MLAGent-Bench, A-Lab / GNoME Materials Discovery Lineage, Coscientist
2024	7	DiscoveryBench, Sci-Code, The AI Scientist, SciAgents, DS Bench, CORE-Bench, MLE-bench
2025	16	Agent Laboratory, OpenAI Deep Research, AI Co-Scientist / Co-Scientist, BixBench, ResearchBench, PaperBench, The AI Scientist-v2, Deep Research Bench / RetroSearch, FutureHouse Platform Agents, AlphaEvolve, Robin, DeepResearch Bench, AstaBench, Asta, DeepScientist, AInsteinBench
2026	2	MMDeepResearch-Bench, AIRS-Bench

5 Capability Patterns Across the Scientific Lifecycle

Across the corpus, five recurring capability patterns appear. First, **research synthesis** systems search, screen, summarize, and cite literature. Second, **hypothesis systems** propose candidate explanations, inspirations, or research plans. Third, **computational research agents** write and run code for machine-learning or data-science tasks. Fourth, **domain discovery systems** couple LLM reasoning with domain tools, laboratory automation, or external simulators. Fifth, **evaluation systems** define tasks,

rubrics, and environments that convert qualitative research behavior into comparable scores.

The most important design tension is breadth versus validation. Broad agents cover more of the lifecycle but are harder to evaluate rigorously. Domain systems are narrower, but they can sometimes close the loop with real experiments, physical instruments, or task-specific metrics. Benchmarks sit between those extremes: they enable comparison, but they can encourage optimization for benchmark-specific behavior rather than genuine discovery.

6 Evaluation Benchmarks and What They Measure

The benchmark layer is the most developed part of the corpus. It covers ML experimentation, paper replication, computational reproducibility, data-driven discovery, scientific coding, bioinformatics, deep research, multimodal report generation, and scientific software maintenance. These benchmarks do not measure the same construct. Some reward code that runs; others reward literature-grounded reports; others test whether an agent can reproduce a paper or maintain a scientific codebase.

Benchmark	Release	Target autonomy	Evaluates
AInsteinBench	2025-12-24	L2-L3	scientific repository maintenance, scientific computing code agents, real PR tasks, feature implementation
AIRS-Bench	2026-02-09	L3-L4	full ML research lifecycle, idea generation, experiment analysis, iterative refinement
AstaBench	2025-08	L1-L3	scientific research agents, paper finding, scholarly question answering, tool use
BixBench	2025-02-28		bioinformatics, biological data analysis, multi-step dataset exploration, open-ended computational biology reasoning

Benchmark	Release	Target autonomy	Evaluates
CORE-Bench	2024-09-17	L2-L3	computational reproducibility, research artifact execution, code reproducibility
Deep Research Bench / RetroSearch	2025-05	L2	web research agents, frozen-web retrieval, tool use, hallucination
DeepResearch Bench	2025-06	L2	deep research agents, citation quality, retrieval accuracy, long-form report quality
DiscoveryBench	2024-07-01	L2-L3	data-driven discovery, hypothesis generation, multi-step reasoning, data semantics
DSBench	2024-09-12		data science, data analysis, data modeling, spreadsheet and Kaggle-style task solving
MLAgentBench	2023-10-05		ML experimentation, code execution, model improvement, long-term planning
MLE-bench	2024-10-09	L2-L3	ML engineering, Kaggle-style competition performance, agent scaffolds, resource scaling behavior
MMDeepResearch-Bench	2026-01	L2	multimodal deep research, image-text evidence use, citation grounding, report integrity

Benchmark	Release	Target autonomy	Evaluates
PaperBench	2025-04-02		paper-to-code replication, AI research engineering, long-horizon agent execution, autonomous experiment execution
ResearchBench	2025-03-27	L2-L3	scientific discovery, inspiration-based reasoning, hypothesis generation, literature-grounded ideation
SciCode	2024-07-18		scientific coding, research problem solving, natural-science computation, code synthesis

A key conclusion is that benchmark plurality is necessary. PaperBench and MLE-bench stress research engineering; DiscoveryBench and ResearchBench stress hypothesis generation; CORE-Bench and AInsteinBench stress reproducibility and scientific software; DeepResearch Bench and RetroSearch stress source-grounded reporting; BixBench and SciCode add domain-specific scientific reasoning. A single leaderboard cannot represent this ecosystem without hiding major differences in task definition.

7 Autonomy, Control, and Human Roles

The records show a consistent pattern: the more a system approaches end-to-end autonomy, the more it needs explicit boundaries. Boundaries include benchmark task definitions, sandboxed execution, human approval, curated datasets, laboratory constraints, citation requirements, or post-hoc expert review. Human scientists remain central as task framers, safety reviewers, domain validators, and judges of novelty.

The practical autonomy ladder in this corpus is therefore relational. L1 systems support human search and synthesis. L2 systems chain tools within a bounded workflow. L3 systems propose and iterate on hypotheses or experiments while relying on human gates. L4 systems execute a full loop in a bounded domain. The evidence base does not yet support treating any of these systems as open-ended autonomous scientists.

8 System Relationships and Ecosystem Structure

Several relationships cut across the records. The AI Scientist line motivates end-to-end machine-learning research loops and connects naturally to PaperBench, MLE-bench, MAgentBench, and AIRS-Bench. Chemistry and materials systems form a second lineage through ChemCrow, Coscientist, A-Lab/GNoME, and related domain agents. Research products such as Elicit, Consensus, OpenAI Deep Research, Asta, and FutureHouse agents form a product layer that is less autonomous than the research prototypes but more likely to be used by working scientists.

This structure matters because progress in one layer does not automatically transfer to another. A benchmark-winning coding agent may not be a reliable literature reviewer. A strong deep-research product may still lack experimental execution. A laboratory agent that succeeds in a narrow chemistry protocol may not generalize to open-ended theory formation.

9 Risks, Limitations, and Open Problems

Three risks recur. The first is **scientific integrity risk**: systems can produce plausible but unsupported novelty claims, fabricated or weak citations, and papers that look complete despite fragile evidence. The second is **evaluation risk**: benchmark performance may reflect task-specific adaptation, judge-model bias, or contamination rather than robust research competence. The third is **operational risk**: agents with code execution, web access, chemical planning, or laboratory control require stronger safety and audit mechanisms than ordinary search tools.

The most important open problems are therefore not only model capability problems. They include provenance tracking, benchmark contamination control, reproducible execution logs, human-in-the-loop review design, domain-specific safety policies, and better separation between report quality and discovery quality.

10 Future Directions

Future AI Scientist work should move from impressive demonstrations toward controlled evidence. Promising directions include modular evaluation of lifecycle stages, benchmark suites that combine code execution with source-grounded reporting, domain tasks with expert validation, and systems that expose their reasoning, tool calls, failures, and uncertainty. The field also needs negative results: failed hypotheses, invalid experiments, and non-reproducible agent outputs should be preserved rather than filtered away.

11 Corpus Census

The full corpus is listed below to preserve the total-count requirement and make the synthesis auditable.

#	Item		Category	Release	Autonomy	Domains
1	A-Lab GNoME Materials Discovery Lineage	/	Domain scientific- discovery agents	2023-11	L4	materials science, inorganic chemistry
2	Agent Labora- tory		Research workflow agents	2025-01-08	L3	machine learning, general com- putational research
3	AI Scientist Co-Scientist	Co- /	Hypothesis and research- planning systems	2025-02-19	L3	biomedicine, biology, gen- eral science
4	AInsteinBench		Evaluation benchmarks and datasets	2025-12-24	L2-L3	scientific computing, quantum chemistry, quantum computing, molecular dynamics
5	AIRS-Bench		Evaluation benchmarks and datasets	2026-02-09	L3-L4	machine learning, nat- ural language processing, mathematics, coding
6	AlphaEvolve		Domain scientific- discovery agents	2025-05-14	L3	algorithms, mathematics, systems opti- mization, AI infrastructure
7	Asta		Science agent product suites	2025-08-26	L2	general sci- ence

#	Item	Category	Release	Autonomy	Domains
8	AstaBench	Evaluation benchmarks and datasets	2025-08	L1-L3	scientific research, literature un- derstanding, code exe- cution, data analysis
9	BixBench	Evaluation benchmarks and datasets	2025-02-28		bioinformatics, computational biology, genomics, transcrip- tomics
10	ChemCrow	Domain scientific- discovery agents	2023-04-11	L2	chemistry
11	Consensus	Literature- review prod- ucts	2021	L1	general academic research
12	CORE-Bench	Evaluation benchmarks and datasets	2024-09-17	L2-L3	computational reproducibil- ity, computer science, so- cial science, medicine
13	Coscientist	Domain scientific- discovery agents	2023-12-20	L4	chemistry

#	Item	Category	Release	Autonomy	Domains
14	Deep Re- search Bench / RetroSearch	Evaluation benchmarks and datasets	2025-05	L2	web research, open-web retrieval, multi-domain factual syn- thesis, tool- using agents
15	DeepResearch Bench	Evaluation benchmarks and datasets	2025-06	L2	deep research, web research, citation- rich report generation, science and technology
16	DeepScientist	End-to-end AI Scientist sys- tems	2025-09-30	L4	machine learning, scientific tasks
17	DiscoveryBench	Evaluation benchmarks and datasets	2024-07-01	L2-L3	sociology, biology, humanities, economics
18	DSBench	Evaluation benchmarks and datasets	2024-09-12		data science, data analysis, data model- ing, tabular reasoning
19	Elicit	Literature- review prod- ucts	2021-08-31	L1	general academic research, systematic reviews
20	FutureHouse Platform Agents	Science agent product suites	2025-05-01	L2	biology, chemistry, biomedicine

#	Item	Category	Release	Autonomy	Domains
21	MLAgentBench	Evaluation benchmarks and datasets	2023-10-05		machine learning experimentation, ML engineering, agentic code execution, model improvement
22	MLE-bench	Evaluation benchmarks and datasets	2024-10-09	L2-L3	machine learning engineering, Kaggle-style competitions, applied machine learning
23	MMDeepResearch Bench	Evaluation benchmarks and datasets	2026-01	L2	multimodal deep research, image-text evidence synthesis, long-form report generation, general research
24	OpenAI Deep Research	Deep-research products	2025-02-02	L2	finance, science, policy, engineering
25	PaperBench	Evaluation benchmarks and datasets	2025-04-02		AI research engineering, machine learning paper replication, long-horizon agent execution

#	Item	Category	Release	Autonomy	Domains
26	ResearchBench	Evaluation benchmarks and datasets	2025-03-27	L2-L3	general scientific discovery, natural language processing, hypothesis generation, literature-grounded research
27	Robin	Domain scientific-discovery agents	2025-05-19	L3	biology, biomedicine
28	SciAgents	Domain scientific-discovery agents	2024-09-09	L3	materials science, bioinspired materials
29	SciCode	Evaluation benchmarks and datasets	2024-07-18		scientific coding, mathematics, physics, chemistry
30	The AI Scientist	End-to-end AI Scientist systems	2024-08-12	L4	machine learning
31	The AI Scientist-v2	End-to-end AI Scientist systems	2025-04-07	L4	machine learning

12 Key Source Links

Item	Primary or official source
A-Lab / GNoME Materials Discovery Lineage	https://www.nature.com/articles/s41586-023-06734-w
Agent Laboratory	https://arxiv.org/abs/2501.04227
AI Co-Scientist / Co-Scientist	https://www.nature.com/articles/s41586-026-10644-y
AInsteinBench	https://arxiv.org/abs/2512.21373
AIRS-Bench	https://arxiv.org/abs/2602.06855
AlphaEvolve	https://arxiv.org/abs/2506.13131
Asta	https://arxiv.org/abs/2510.21652
AstaBench	https://arxiv.org/abs/2510.21652
BixBench	https://arxiv.org/abs/2503.00096
ChemCrow	https://arxiv.org/abs/2304.05376
CORE-Bench	https://arxiv.org/abs/2409.11363
Coscientist	https://www.nature.com/articles/s41586-023-06792-0
Deep Research Bench / RetroSearch	https://arxiv.org/abs/2506.06287
DeepResearch Bench	https://arxiv.org/abs/2506.11763

Item	Primary or official source
DeepScientist	https://arxiv.org/abs/2509.26603
DiscoveryBench	https://arxiv.org/abs/2407.01725
DSBench	https://arxiv.org/abs/2409.07703
Elicit	https://elicit.com/
MLAgentBench	https://arxiv.org/abs/2310.03302
MLE-bench	https://arxiv.org/abs/2410.07095
MMDeepResearch-Bench	https://arxiv.org/abs/2601.12346
OpenAI Deep Research	https://openai.com/index/introducing-deep-research/
PaperBench	https://arxiv.org/abs/2504.01848
ResearchBench	https://arxiv.org/abs/2503.21248
Robin	https://arxiv.org/abs/2505.13400
SciAgents	https://arxiv.org/abs/2409.05556
SciCode	https://arxiv.org/abs/2407.13168
The AI Scientist	https://arxiv.org/abs/2408.06292
The AI Scientist-v2	https://arxiv.org/abs/2504.08066